

Project Design Phase
Proposed Solution Template

Date	03 October 2026
Team ID	SWTID-2026-5823
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Proposed Solution Template:

Proposed Solution

Script-Controlled ACL – Restrict Record Access Based on Field Value

S.No.	Parameter	Description
1	Problem Statement (Problem to be solved)	Users accessing the Institution Details table should not be able to view or modify records outside their authorized branch. A branch-based access mechanism is required so that EEE users can view only records whose Branch field is EEE, while administrators retain full access. Create, write and delete operations must also be controlled according to assigned roles.
2	Idea / Solution description	Implement ServiceNow record-level Access Control Lists (ACLs) on the custom u_institution_details table. Create custom roles bb1, bb2, bb3 and bb4. The READ ACL requires bb1, uses the condition Branch is EEE, and includes an advanced script that grants administrators full access and permits bb1 users. Separate CREATE, WRITE and DELETE ACLs require bb2, bb3 and bb4 respectively. Users are validated through impersonation and list/form access testing.
3	Novelty / Uniqueness	The solution combines role-based authorization with a record-field condition. Instead of granting access to the complete table based only on a role, the READ ACL evaluates the Branch value so that EEE users are restricted to EEE records. Different roles are also used to control the lifecycle operations independently: create, write and delete.
4	Social Impact / Customer Satisfaction	The solution protects student and institution information from unauthorized access and reduces accidental modification or deletion. Authorized users receive access appropriate to their responsibilities, while administrators can continue to manage all records. This improves trust, data confidentiality and operational control within the ServiceNow environment.
5	Business Model (Revenue Model)	The proposed solution is primarily a ServiceNow security and access-control implementation rather than a standalone revenue-generating product. It can be delivered as part of ServiceNow implementation, customization, security configuration or consulting services. Organizations may use the approach to reduce data exposure and administrative risk.
6	Scalability of the Solution	The ACL-based approach can be extended to additional branches, departments, roles and tables. More field-level or record-level conditions can be added as requirements grow. The same pattern can control other CRUD operations and can be applied across ServiceNow applications while preserving centralized authorization and auditability.